EDGAR: Evidence-Driven Guarded Agentic Retrieval for Video Events

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Abstract. Video event retrieval requires finding a precise moment, answering a question about it, or locating an ordered sequence of events. Similar scenes and agent confidence do not establish when to stop searching. We present EDGAR (Evidence-Driven Guarded Agentic Retrieval for Video Events), developed by team APCS ⊕ CNTN for the Ho Chi Minh City AI Challenge 2026. EDGAR combines multimodal search with two tool-using agents. An evidence board records observations and their sources; a decision model assesses support and selects actions; a code-level guard checks stopping conditions. We evaluate six configurations on 86 test queries, with a follow-up ablation of final verification after parallel search. In this follow-up, EDGAR with holistic verification achieves 58.1% strict top-1 success (correct timing and task-specific answers), versus 47.7% for always-two agents and 50.0% with final verification added. Its advantage over the latter remains inconclusive (exact McNemar $p = 0.092$). Median latency rises from 53.7 to 75.8 s relative to always-two agents, with negligible reduction in agent calls. The evaluation exposes an accuracy–latency trade-off; the guard's individual contribution remains unresolved.

Keywords: Video event retrieval · LLM agents · Fast and slow control · Decision models · Evidence verification · Multimodal retrieval

1 Introduction

The Ho Chi Minh City AI Challenge 2026 (AIC 2026) [12] asks for a described moment (KIS), a moment and answer (QA), or an ordered event chain (TRAKE). Joint embeddings retrieve candidates quickly [5, 29], but descriptions combine actions, appearance, text and speech across repeated broadcast scenes. Our retrieval baseline ranks the correct video first on 45.3% of queries, yet strict R@1 is 15.1%.

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Tool-using agents search, inspect frames and refine queries [25, 28, 40, 43]. Running both agents costs tens of seconds; their confidence is uncalibrated, and disagreement alone does not justify further search.

Team APCS ⊕ CNTN developed EDGAR for AIC 2026. With fast and slow control [21, 27], System-1 retrieves, a decision model [37] assesses evidence and selects typed actions, and System-2 agents inspect media. The guard separates evidence-sufficient stops from limits; these roles imply no latency reduction. Our contributions are:

- **Traceable system design**: an evidence board, typed decisions and a stop guard coordinate multimodal retrieval and two agents sharing ten Model Context Protocol (MCP) tools [1].
- **Evaluation and diagnostics**: comparisons and analysis of fixed evidence distinguish final ranking from controller behaviour. Group-level metrics and cumulative hints characterise coverage and retrieval with partial queries.
- **Deployment experience**: official AIC 2026 results and operator practices document the console in operator-assisted use alongside agents.

2 Related Work

Interactive video retrieval. VBS systems combine embeddings and browsing [29, 34], spatial and temporal queries [8, 20, 22], answer grounding [36], fusion [23] and graded autonomy [16]. AIC systems use dynamic programming and visual verification [15, 31]; agents also mediate video and lifelog search [17, 38].

LLM agents for video. Video agents inspect frames [40], select tools [43], ground temporal reasoning [25, 28] and coordinate retrieval [41]. Jäckl et al. [19] combine shared search state, structured planner actions and deterministic validation. Building on interleaved reasoning and acting [42], EDGAR separates typed evidence assessment and action selection from an executable stop guard.

Fast and slow control and routing. SwiftSage pairs fast policies with slow planning [27]; FrugalGPT and RouteLLM study model allocation for cost and quality [7, 32]. EDGAR uses typed support probabilities and a separate guard instead of relying on free-form judge rationales [45]. We examine the resulting accuracy–latency trade-off and score reliability [14].

3 System Overview

The system (Fig. 1) indexes 1,478 broadcasts plus batch-two news, 298 traffic-camera and 12 cycling recordings, with organiser (382,299 frames) and InfoShot++ (2,212,383 frames) profiles. Our InfoShot++ sampling algorithm extends InfoShot [44] with multiple unusual within-shot candidates, native-frame-rate refinement around text/motion changes, and temporal gap filling. Its benefit is not ablated. Milvus [39] stores PE-Core [5], Qwen3-VL [26] and GLAP [11] vectors;

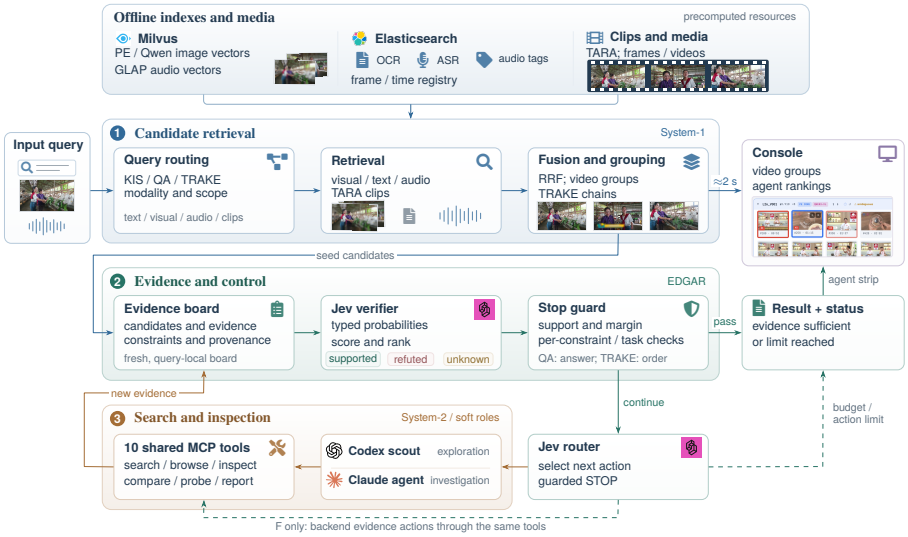


Fig. 1. EDGAR architecture: retrieval, typed verification, guarded stopping and agent/tool actions. Dashed paths denote F-only macros and limit termination; evidence returns to the query-local board. Media and UI excerpts are illustrative.

Elasticsearch stores OCR [18], WhisperX-aligned Vietnamese ASR [4,24], audio tags [10] and frame/time mappings. TARA [3] indexes 496,155 clips across core and batch-two videos (partial cycling coverage); traffic windows span 4/8/16 s. Retrieval returns groups in about two seconds; a cancellable sidecar streams agent rankings. Section 6 gives the benchmark setup.

4 System-1 Retrieval and Group-by-Video Ranking

A heuristic parser routes each query to visual (English translation), OCR (exact and fuzzy Vietnamese phrases), speech, audio and clip channels, and infers folder-scope filters from the topic. Channel c returns a frame ranking L_c, and weighted reciprocal rank fusion [9] scores frame f by $R(f) = \sum_{c:f\in L_c} w_c/(60+r_c(f))$ with ranks $r_c(f) \geq 1$; cosine scores from different embedding spaces are never added.

Group-by-video ranking. The console groups frames for playback (Fig. 2). For video v, returned frames F_v, best frame $f_v^\star$ and $R^\star = \max_f R(f)$,

$$S(v) = 0.75\,\frac{R(f_v^\star)}{R^\star} + 0.15\,\frac{\text{mean}_{f\in\text{top3}(F_v)}\,R(f)}{R^\star} + 0.10\,\min\!\left(1, \frac{n_{30}(v)-1}{4}\right), \quad (1)$$

where $n_{30}(v)$ counts returned frames within 30 s of $f_v^\star$, including itself. These heuristic weights are unablated; top-frame gaps over 30 s flag ambiguity. With TARA, video-level RRF over channels and clip scales replaces Eq. (1).



Fig. 2. Console workflow: query and search controls (left), grouped video results (centre), and frame inspection, evidence and submission identifiers (right).

TRAKE. A shared context (subject, clothing, scene) is added to every event description, events are retrieved separately, temporal non-maximum suppression keeps distinct moments, and an exact dynamic program selects the maximum-weight chain with increasing times in each video. Videos are ranked by coverage, then chain quality.

5 EDGAR: Evidence-Driven Guarded Agentic Retrieval

Evidence board. Each query owns a fresh board $\mathcal{B}_t$; no previous hint or reasoning enters it. A deterministic compiler retains the query (weight 2), extracts clauses and attribute, relation or negation spans (weight 1), and adds TRAKE events (weight 1.5) or a QA answer constraint (weight 2). Evidence is de-duplicated by kind and source. Visual reports require prior frame loading, recording inspection rather than guaranteed truth. Reports from the same media need not be independent. Agent confidences order proposals but are not used as probabilities.

Typed verification. Each step assesses five candidates: agents' leading proposals, the verified leader, then distinct fused-ranking moments. TRAKE allows up to 12 to cover the leading video's events. Jev 1.13 assesses textual evidence, not images. For candidate moment x and constraint c, it returns probabilities over $\{\textit{supported}, \textit{refuted}, \textit{unknown}\}$, without inferring visual details from scores or confidence. "Non-generative" describes this interface: probabilities replace free-form plans or queries. Without evidence, code sets $p^{\text{sup}} = 0$. The support score is

$$s(x) = \exp\Big(\sum_{c \in \mathcal{C}(x)} w_c \log \max\big(\epsilon, p_c^{\text{sup}}(x)\big) \Big/ \sum_{c \in \mathcal{C}(x)} w_c \Big), \tag{2}$$

a weighted geometric mean over the assessed constraints $\mathcal{C}(x)$, with $\epsilon = 10^{-9}$. It penalises weak support but is not a calibrated joint probability of correctness;

Algorithm 1 EDGAR controller for one query (6 steps, 40 tool calls, 240 s).

Input: query q, task type, System-1 result R_0

$\mathcal{C}_q \leftarrow$ Compile(q); $\mathcal{B}_0 \leftarrow$ Seed($R_0, \mathcal{C}_q$); $\mathcal{I} \leftarrow \emptyset$ ▷ fresh, query-local
for $t = 0, 1, \ldots$ **while** budget remains **do**
  JevVerify($\mathcal{B}_t$); Publish(ranking of $\mathcal{B}_t$)
  if guard (3) holds **then return** ▷ evidence sufficient
  $\mathcal{A}_t \leftarrow \{\texttt{STOP}\} \cup \{\text{calls to agents} \notin \mathcal{I}\} \cup \{\text{untried evidence actions in mode F}\}$
  $a_t \leftarrow$ JevRoute($\mathcal{B}_t, \mathcal{A}_t$); **if** $a_t = \texttt{STOP}$ **then** $a_t \leftarrow$ next escalation in $\mathcal{A}_t$
  if $a_t = \texttt{STOP}$ **then return** ▷ actions exhausted
  $\mathcal{B}_{t+1} \leftarrow$ Execute($a_t, \mathcal{B}_t$); $\mathcal{I} \leftarrow \mathcal{I} \cup$ agents in a_t

full-query and decomposed constraints can overlap. *Holistic* mode (E) assesses the query and, for QA, answer constraints; *constraint-level* mode (F) assesses all applicable compiled constraints and enables evidence actions. Ranking promotes candidates for which *supported* is largest on every assessed constraint, demotes those with any *refuted* maximum, and otherwise preserves System-1/agent order. Thus *unknown* is abstention; promotion need not satisfy the stop threshold.

Routing and the stop guard. While the guard fails, Jev chooses the next action from $\mathcal{A}_t \subseteq \{$`STOP`, `CALL_CODEX`, `CALL_CLAUDE`, `CALL_BOTH`, `SEARCH_MORE`, `INSPECT_TOP1`, `COMPARE_TOP2`$\}$, each described in one sentence; the first request also classifies modality and query structure. Each agent runs at most once. The evidence actions (mode F only) are backend macros: a focused search for the least-supported constraint, or OCR and speech around the top two candidates. Agent disagreement enters the state as a signal, and the router is told it alone does not justify another agent. Evidence-sufficient stopping requires, for top candidate x_1,

$$\begin{gathered} s(x_1) \geq \tau, \qquad s(x_1) - \max_{x' \in \mathrm{comp}(x_1)} s(x') \geq \delta, \\ p_c^{\mathrm{sup}}(x_1) > \max\!\left(\tfrac{1}{2},\ p_c^{\mathrm{ref}}(x_1),\ p_c^{\mathrm{unk}}(x_1)\right)\ \forall c \in \mathcal{C}(x_1), \end{gathered} \tag{3}$$

where $\mathrm{comp}(x_1)$ contains scored competitors outside the same moment (≤ 8 s apart in one video); TRAKE compares other videos. The maximum is zero when this set is empty. Thus the margin concerns assessed competitors, not unassessed candidates. QA needs an answer; TRAKE needs supported events in increasing time. We use raw-score thresholds $\tau = 0.9$, $\delta = 0.1$. A rejected `STOP` triggers the next escalation (Codex, Claude, inspect, search, compare). Budget or action exhaustion returns the current ranking with a distinct termination status (Algorithm 1). Passing the guard is a policy decision, not a correctness guarantee.

System-2 agents and tools. Codex CLI with GPT-6.1 Sol [33] and Claude Code with Opus 5.5 [2] use high reasoning effort and fast mode in fresh processes without shell/file tools. Both share ten MCP tools for retrieval, corpus browsing, local frame/text inspection, candidate comparison and reporting (inventory in the repository, Sect. 6). Soft roles favour exploration or counterexample inspection without restricting capabilities. Prompts describe the corpus and include a bounded board snapshot marked as untrusted data.

6 Experimental Setup

Benchmark. We assembled 111 queries from the organisers' AIC 2026 query workbooks and final-round submissions. Ground truth combines DRES-accepted submissions [35] with owner-reviewed annotations, each checked against the source frames or clips: moments for all tasks, accepted answers for QA and ordered per-event alternatives for TRAKE. Queries are split by content hash and grouped by target video, so no target video appears in both DEV (25) and TEST (86). TEST has 59 T-KIS, 19 QA and 8 TRAKE queries over 87 target videos, using InfoShot++ (46 queries) or organiser (40) profiles. DEV fixed the policy; its fitted calibration was not adopted, so all configurations use raw probabilities.

Configurations. All six share System-1, soft roles and tools. **A**: retrieval only. **B**: always Codex. **C**: Codex and Claude in parallel. **D**: Jev holistic reranking without agents. **E**: EDGAR with holistic verification. **F**: EDGAR with constraint-level verification and evidence actions. B/C use fixed schedules and RRF fusion; E/F add verification, routing and guarded stopping. D assesses retrieval evidence alone; E–F changes granularity and evidence actions. System-1 uses PE-Core on both profiles, heuristic parsing with VI→EN translation, no LLM rewriting/reranking, and 150 frames in at most 30 videos; OCR/speech/audio/TARA activate for 18/11/9/5 TEST queries.

Metrics. Strict R@k and MRR require an accepted moment (point/interval, ± 1 s); QA additionally requires a normalised accepted answer, TRAKE a complete ordered sequence in one video. Ranks count frames for T-KIS/QA and videos with proposed sequences for TRAKE. *Video* R@k checks target identity among the first k distinct videos. *Group-moment* R@k requires a returned matching moment in those groups, ignoring QA answers but retaining complete TRAKE sequences. Videos follow first occurrence in the ranking. Rescue@1 measures agent-provenance strict success when the retrieval seed misses top-1. We record latency, calls, tokens, ECE (10 bins) and Brier score [6] over final scored non-TRAKE candidates, comparing support scores with strict success rather than constraint-support labels. Plots use five bins; candidates are not independent query samples.

Protocol. Fresh query–configuration runs use seeded random variant order. The initial round retains 516 completed TEST runs; the follow-up has 258 on the same 86 queries, with unchanged policy, prompts, tools, retrieval and budgets. C+V adds E's final holistic verification to C, saves pre-verification rankings and disables routing/adaptive stopping. TEST data are reused without new DEV tuning. Quota-interrupted attempts restart fresh: 13 interruptions add 314.6 s and 25 agent calls outside completed-run latency; completed runs have no errors/fallback. Percentile bootstraps [13] resample queries: 2,000 draws for initial intervals, 10,000 paired draws (seed 2026) for contrasts. Exact McNemar tests [30] accompany exploratory comparisons without multiplicity correction.

Table 1. Initial TEST round (N=86; tolerance 1 s). R@1: bootstrap 95% CI, 2,000 draws; Resc.: Rescue@1, 73 eligible runs/variant; p50: median latency; Agents: invocations/query; kTok: mean agent and decision tokens in thousands.

	Configuration	R@1 [95% CI]	R@5	MRR	Resc.	p50 (s)	Agents	kTok
A	Retrieval only	15.1 [8.1, 23.3]	33.7	.233	–	1.6	0	0
D	Jev rerank	15.1 [8.1, 23.3]	33.7	.233	0.0	2.1	0	3
B	Always Codex	39.5 [29.1, 50.0]	68.6	.529	28.8	50.5	1.00	230
C	Always both	51.2 [40.7, 61.6]	**72.1**	.612	42.5	51.2	2.00	323
E	EDGAR, holistic	**58.1** [47.7, 68.6]	67.4	**.630**	**52.1**	72.6	1.95	316
F	EDGAR, constraint	52.3 [41.9, 62.8]	64.0	.572	45.2	84.2	1.99	416

7 Results

7.1 Initial Six-Configuration Comparison

E has the highest observed strict R@1 in Table 1 (58.1%). E improves on B by 18.6 points (95% CI [9.3, 27.9]; McNemar $p < 0.001$). Its E–C difference is inconclusive (+7.0, [−2.3, 16.3]; 11 E-only and 5 C-only successes, $p = 0.21$). C has the highest R@5 (72.1% vs. 67.4%).

D equals A: all 525 assessments were *unknown*, preserving System-1 order. This checks abstention on retrieval evidence, not verification after agent inspection. F has lower observed R@1 than E (−5.8 points, [−15.1, 3.5]), but the contrast is inconclusive and changes both granularity and evidence actions. F leads initial TRAKE R@1 (50.0% vs. E's 37.5% and C's 25.0%; eight queries only).

7.2 Live C/C+V/E Ablation

Table 2 tests final verification after always-two acquisition. All 86 C+V traces show both agents, one final verification and no routing. C+V–C and E–C+V remain inconclusive. E–C is positive in this round (unadjusted $p = 0.035$), unlike the initial round, but does not establish a gain over the C+V control.

C and C+V share an acquisition policy but have independent trajectories. Comparing E with C+V changes scheduling, verification, board context and stopping together; neither guard nor router is isolated.

Task-level variability. Follow-up C/C+V/E R@1 is 57.6/61.0/74.6% for T-KIS, 26.3/26.3/31.6% for QA and 25.0/25.0/0.0% for TRAKE. E changes from 39/59 to 44/59 T-KIS successes, 8/19 to 6/19 QA and 3/8 to 0/8 TRAKE. The identical total (50/86, 58.1%) masks this variability. E's TRAKE target-video R@1 falls from 6/8 to 3/8, and localised events from 15/28 to 6/28 within ±1 s in its top video. Two queries were never solved; partial chains receive no strict credit.

Final verification on fixed evidence. On identical C+V candidates and answers, final verification raises R@1 from 40/86 (46.5%) to 43/86 (50.0%): four repairs and one harm. This same-evidence difference is inconclusive (Table 2).

Table 2. Follow-up TEST round (N=86); recall in percent. Lower panel: paired R@1 contrasts, 95% bootstrap CIs (10,000 draws), exact unadjusted McNemar p and discordant win/loss counts. "Pre": C+V's own ranking before verification.

	R@1	R@5	MRR	p50 (s)	p95 (s)	Agents	Jev calls
C	47.7	**69.8**	.579	53.7	103.2	2.00	0.00
C+V	50.0	66.3	.581	55.6	98.4	2.00	1.00
E	**58.1**	66.3	**.620**	75.8	198.9	1.95	4.91

Contrast	ΔR@1 (pp) [95% CI]	Win/loss	p
C+V − C	+2.3 [−3.5, 9.3]	5/3	.727
E − C+V	+8.1 [0.0, 16.3]	10/3	.092
E − C	+10.5 [2.3, 18.6]	12/3	.035
C+V − Pre	+3.5 [−1.2, 9.3]	4/1	.375

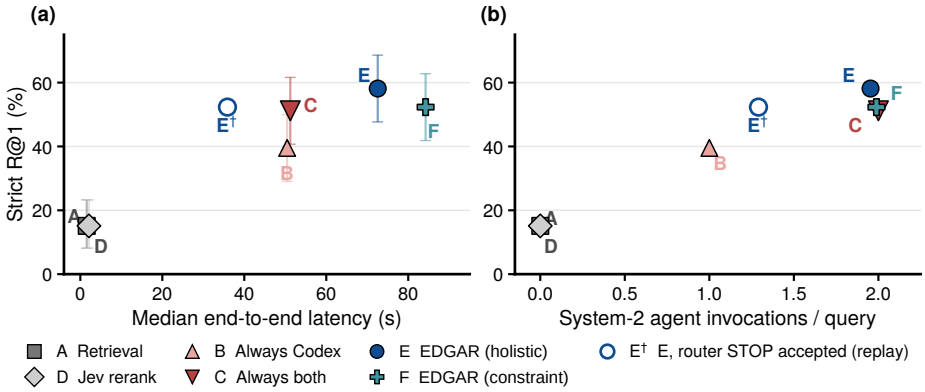


Fig. 3. Initial-round strict R@1 versus latency (a; 95% bootstrap CIs) and agent calls (b). E† is an offline first-`STOP` replay, not a deployed configuration (Sect. 7.3).

In the initial round, re-sorting final candidates by logged RRF, first-seen time and frame ID reproduces all 344 A/B/C/D rankings. It gives E 51/86 versus verified 50/86 and F 48/86 versus 45/86 (95% CIs $[-5.8, 2.3]$ and $[-9.3, 1.2]$ points). No reliable final-ranking benefit is demonstrated; earlier verification may still affect acquisition and agent context.

7.3 Accuracy, Latency and Stopping: Router versus Guard

In the initial round, EDGAR did not save computation (Fig. 3): E invoked 1.95 agents per query, and its median latency (72.6 s; p95 171 s) exceeds C's (51.2 s; p95 94 s). The router sent the investigator first in 83/86 queries; sequential escalation adds agent times. Jev averaged 4.9 calls and 3.2 s/query (Fig. 4a). The follow-up round confirms this pattern (Table 2); E/C+V/C use 327k/344k/346k tokens, a workload proxy rather than complete cost.

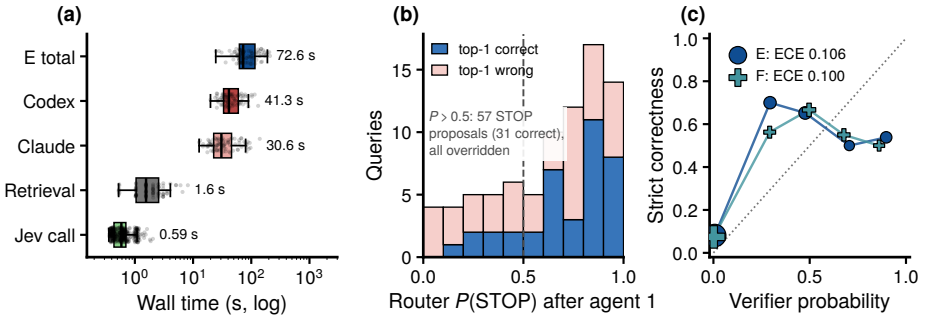


Fig. 4. Initial controller diagnostics. (a) E wall time per layer (log scale). (b) Router $P(\text{STOP})$ after agent 1 by top-1 correctness; every proposal above 0.5 was overridden. (c) E/F verifier reliability (marker area $\propto$ count).

After agent 1, E proposed `STOP` above probability 0.5 in 57/82 decisions (median 0.73): 31/57 top-1 results were correct, versus 7/25 otherwise (Fig. 4b). The guard blocked all 57; nine runs ended evidence-sufficient. Offline E† retains the ranking, calls and time before the first router `STOP`, else the final outcome: 1.29 agents/query, 35.9 s median, 52.3% R@1, versus E's 1.95, 72.6 s, 58.1%. This post-hoc TEST replay establishes neither live latency nor equivalence to C. A DEV-only replay accepts post-agent `STOP` when $P(\texttt{STOP}) \geq \theta$: the smallest $\theta \in \{0.50, 0.55, \ldots, 0.95\}$ preserving DEV R@1 is 0.9. This stops 14 and 15 runs early in the two TEST rounds without changing strict outcomes, reducing agents to 1.79/1.78 and median latency to 70.7/73.6 s from 72.6/75.8; $\theta = 0.7$ yields 51.2% R@1 at 48.1/49.7 s. Speculative parallelism could reduce sequential delay but needs live evaluation.

Support scores imperfectly predict strict correctness (ECE 0.106, Brier 0.112; Fig. 4c): only 7/13 candidates scoring at least 0.8 were correct. These labels concern exact moments/answers, not constraint support; the 13 candidates are not the set of runs passing the full stop guard.

7.4 Group-by-Video Evaluation

Figure 5 separates exact-answer accuracy from coverage in returned video groups: strict, group-moment and video R@1 are 15.1/43.0/45.3% for retrieval and 58.1/81.4/86.0% for E, and group-moment R@5 is 86–90% with agents. These same-ranking gaps motivate browsing, not a gain over ungrouped retrieval. For QA, E's 89.5% group-moment versus 42.1% strict R@1 mixes exact-frame selection and answer extraction; neither is isolated.

7.5 Cumulative-Hint Evaluation

Twenty-four TEST queries (12 T-KIS, 12 QA) have three or four reviewed hints whose concatenation equals the full query. Each prefix h_1, $h_1{+}h_2$, … is an *independent* fresh run without cross-hint memory; we compare A, C and F. With

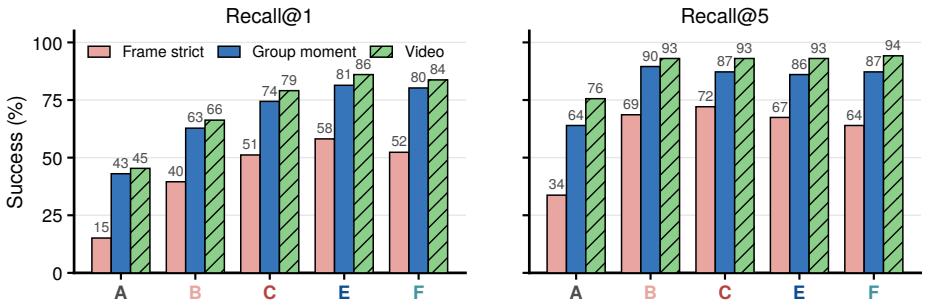


Fig. 5. Initial strict, group-moment and video R@k (D=A; metrics in Sect. 6). This compares metrics, not grouping algorithms.

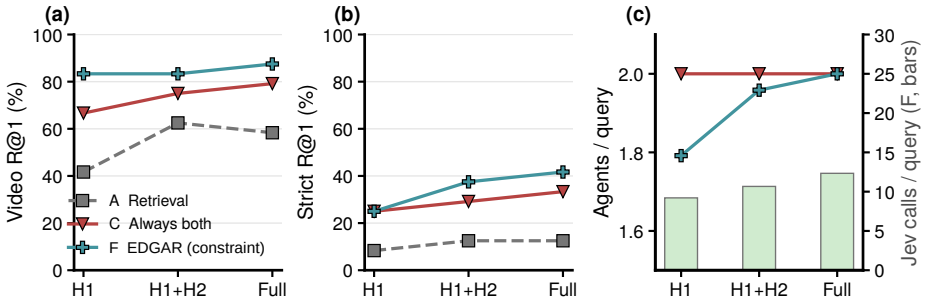


Fig. 6. Cumulative-hint evaluation on 24 TEST queries, each prefix a fresh, independent run. (a) Video R@1. (b) Strict R@1. (c) Agents per query (lines) and Jev calls per query for F (bars).

only the first hint (Fig. 6a), F ranks the target video first for 83.3% of queries, against 66.7% for C (+16.7 points, $[4.2, 33.3]$) and 41.7% for A. On the full query, strict R@1 is 41.7%, 33.3% and 12.5% (F vs. A +29.2, $[12.5, 50.0]$; F vs. C +8.3, $[-8.3, 25.0]$). F uses 1.79/2.0 agents and 9.2/12.3 Jev calls at first/full hint. Its video R@1 succeeds early on 23/24 queries (Table 3); later prefixes may fail.

Qualitative example. Figure 7 traces an initial-round TEST query through E. Retrieval finds the video but an interview scene. Claude finds the feeding moment; Jev supports it (0.73) and refutes a near-duplicate (0.81). The router proposes `STOP` (0.97), but the guard rejects it ($0.73 < \tau$), triggering Codex's inspection.

8 AIC 2026 Final Round: Results and Practices

Team APCS $\oplus$ CNTN used this console in the AIC 2026 final round [12]. The organisers scored our T-KIS at 1219.6 (15 queries; Outstanding), QA at 988.2 (12; Outstanding), TRAKE at 57.9 (3; Very Good) and V-KIS, not studied here, at 450.0 (5; Very Good); overall performance was Outstanding. TRAKE

Table 3. Hints-to-solve on 24 queries. $h^\star$: first successful prefix (H+1 if unsolved); Early: success before the last hint; Hint saving: $(H - h^\star)/(H - 1)$, 0 if unsolved. These retrospective measures do not imply online stopping or compute savings.

	Video criterion				Strict criterion			
	Solved	Mean $h^\star \downarrow$	Early	Hint saving	Solved	Mean $h^\star \downarrow$	Early	Hint saving
A	19	2.25	.750	.611	6	4.04	.250	.153
C	22	1.67	.875	.792	10	3.42	.375	.319
F	**23**	**1.38**	**.958**	**.889**	**12**	**3.25**	**.458**	**.361**

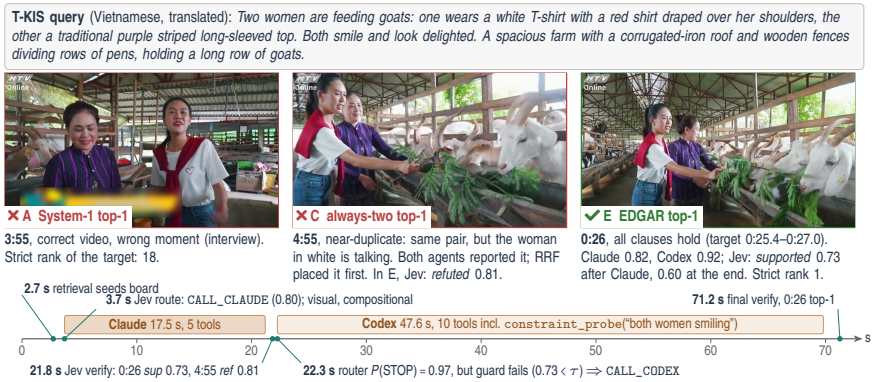


Fig. 7. Initial-round TEST example. Top: top-1 frames of retrieval (A), always-two (C) and EDGAR (E). Bottom: E's logged controller events (teal) and agent work (amber). The left frame's lower-third is blurred.

is also our least stable benchmark task (Sect. 7.2). Of the 30 final-round T-KIS/QA/TRAKE queries, 26 fall in TEST; there, autonomous E ranks the target video first for 24 and 23 in the two rounds but solves 12 strictly (C: 9; A: 3), with QA at 3/12 strictly versus 11–12/12 by video. The live QA score suggests operators completed this step by localising the frame and reading the answer; their effort was not timed.

Operators searched manually while agents ran on the full query in a separate tab (agents off in the manual tab to avoid cancellation), inspecting streamed candidates and frame/OCR/ASR evidence at once. The final-round cases below are operator observations, not matched ablations or measured speedups; LLM rewriting is outside the benchmark.

T-KIS: select clip retrieval. In `tkis-query-23`, a worker towing a bin by motorcycle drops a white bag. TARA returned the correct segment from hint 1, where PE failed; later hints verified it (`N061-V002`, 289.607 s).

T-KIS: use concurrent agents. In `tkis-query-10`, cyclists pass white balloons onto a bridge before six riders appear frontally. Manual PE/TARA localisation

was difficult; a concurrent agent found `S01-V009` at 4542.7 s. Initial A/C rank it second/first; C reports it at 30.4 s, finishing at 62.5 s.

TRAKE: align events manually. The accepted fish-hotpot sequence in `L26_V402` is 91.24, 140.48, 189.12 and 224.24 s. Retain event context and inspect first-contact/last-entry boundaries on paused frames; increasing DP chains cannot guarantee alignment (exact first-frame labels are unavailable), so verify event count, order and identifiers.

9 Discussion and Limitations

Attribution. E versus C+V establishes neither superiority nor equivalence. Controls with Claude alone or matched sequential scheduling are absent. Fixed-evidence analyses concern ranking, not acquisition. Guard overrides do not establish false-stop prevention; the DEV-selected threshold saves few calls.
Evidence. Text-only verification inherits observation errors. Overlapping constraints and geometric aggregation do not establish joint correctness. The margin excludes unassessed candidates and treats moments within 8 s as equivalent, versus 1 s strict tolerance.
Scope. Limitations include one corpus and language, owner-reviewed labels, thematic overlap across video-disjoint splits and bootstraps omitting shared-video dependence. Two TEST rounds cannot establish stability; C+V runs once per query on reused TEST data, TRAKE has eight queries, and hosted models may change. Timed operator studies remain future work.

10 Conclusion

EDGAR is team APCS $\oplus$ CNTN's AIC 2026 system for traceable retrieval and agent control. Its holistic configuration led observed strict R@1, but gains over always-two agents plus final verification remained inconclusive; computation savings were not demonstrated. The team achieved Outstanding overall in the final round. Evidence quality, sequential escalation and router/guard contributions require further evaluation.

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